

Highlights

Interacting effects of wave coupling and bed mobility on Lagrangian transport in a shallow vegetated lagoon

Cicero Martins Jr., Giuseppe Ciruolo, Mauro De Marchis

- Eight model configurations agree on water level but differ by 50% in drifter skill
- Wave coupling and bed mobility each degrade transport skill when applied alone
- Together they improve it, giving the ensemble its two best configurations
- Bed mobility without waves makes every one of the 35 drifters worse
- Surface speed exceeds bed speed twofold at sub-metre mean depth